MIL-C-55134A(EL)
25 October 1967
Superseding
MIL-C-55134(SigC)
21 June 1961

MILITARY SPECIFICATION

CABLES, TELEPHONE (INSIDE)

1. SCOPE

1.1 Scope.- This specification covers the following types of inside telephone cables. (See 6.1).

(a) Type I - 26 pair, No. 24 AWG telephone cable, composed of coated copper conductors, insulated with polyethylene, cabled, marked, bound, wrapped and jacketed with a polyvinylchloride compound, and color coded in accordance with Pair No. 1 through 26 of Table I.

(b) Type II - 14 pair, No. 24 AWG telephone cable, composed of coated copper conductors, insulated with polyethylene, cabled, marked, bound, wrapped and jacketed with a polyvinylchloride compound, and color coded in accordance with Pair No. 1 through 14 of Table I.

(c) Type III - 14 pair, No. 24 AWG telephone cable, composed of coated copper conductors, insulated with polyethylene, cabled, marked, bound, wrapped and jacketed with a polyvinylchloride compound, and color coded in accordance with Pair No. 13 through 26 of Table I.

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

J-C-98	Cable and Wire, Insulated; Methods of Sampling and Testing.
QQ-W-343	Wire, Electrical and Nonelectrical, Copper (Uninsulated).

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MIL-C-12000 Cable, Cord and Wire, Electric, Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions, should be obtained from the procuring agency, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

D-573 Method of test for accelerated aging of vulcanized rubber by the oven method.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania).

3. REQUIREMENTS

3.1 Construction.- Types I, II and III telephone cables, shall be constructed as specified in 1.1 and herein. Only virgin materials shall be used in the manufacture of insulating and jacketing compounds used in the fabrication of cables.

3.2 Conductors.- Each conductor shall consist of 7 strands of coated copper wire, six strands shall be concentrically stranded around the seventh. Each strand shall have a nominal diameter of 0.008 inch. The copper wire shall be in accordance with QQ-W-343, Type C, Class B.

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3.3 Conductor insulation.- Each conductor shall be insulated with high density polyethylene having a nominal wall thickness of 0.010 inch. The minimum wall thickness shall not be less than 0.008 inch. The insulation shall be smooth and free from holes and occlusions. The polyethylene shall be of a recognized electrical grade and be free from foreign materials. It shall contain a suitable antioxidant. Pigments used to color the insulation shall be compatible with the insulation antioxidant.

3.3.1 Physical requirements of insulation before aging.- The insulation shall meet the requirements that follow when tested in accordance with 4.6.

3.3.1.1 Elongation.- The insulation shall have an elongation of the 1 inch or 2 inch gauge marks of not less than 500 percent at rupture when tested in accordance with 4.6.1.

3.3.1.2 Tensile strength.- The insulation shall have a tensile strength of not less than 2500 pounds per square inch of original cross-sectional area at rupture. (See 4.6.2).

3.3.1.3 Compression strength.- The insulation shall have a minimum compression strength of 650 pounds when tested in accordance with 4.6.3.

3.3.1.4 Tubing of insulation.- The specimen shall show no evidence of cracking of the insulation after testing in accordance with 4.6.4.

3.3.1.5 Deformation of insulation.- The decrease in diameter during the period indicated in 4.6.5 shall not exceed 15 percent of the initial reading.

3.3.1.6 Low temperature, insulation.- The specimen shall show no visible evidence of cracking of the insulation when tested and examined in accordance with 4.6.6.

3.3.1.7 Density and melt index of insulation.- The density of the polyethylene shall be between 0.943 and 0.955 and the melt index shall be 0.50 maximum. (See 4.6.7).

3.3.2 Physical requirements of insulation after aging.- The insulation shall meet the requirements that follow when tested in accordance with 4.7.

3.3.2.1 Elongation.- The aged insulation shall have an elongation of the 2 inch gauge marks of not less than 80 percent of the unaged value. (See 4.7.1).

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3.3.2.2 Tensile strength.- The aged insulation shall have a tensile strength of not less than 80 percent of the unaged value, at rupture. (See 4.7.2).

3.3.2.3 Compression strength.- The aged insulation shall have a minimum compression strength of 80 percent of the unaged value when tested in accordance with 4.7.3.

3.3.2.4 Thermal stress cracking.- The specimen shall show no evidence of cracking after testing in accordance with 4.7.4.

3.4 Color code.- The Type I cables shall be fully color coded, shall conform to Table I and shall be readily distinguished. The different colors used for Type II cables shall be those shown for Pair No. 1 through 14 inclusive of Table I and shall be readily distinguished. The different colors used for Type III cables shall be those shown for Pair No. 13 through 26 inclusive of Table I and shall be readily distinguished. The base color shall be an integral part of the insulation. The color band shall be applied using a non-conductive ink over the insulation with a minimum circumferential coverage of 80 percent. The width of the band shall be approximately 1/8 inch. Spacing between the 2 bands shall be approximately 1/16 inch. The color coding of the wire shall repeat every 3/4 inches.

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Table I
Color Code

Pair No.	Layer	Wire		Mate	
		Base Color	Band	Base Color	Band
1	Center	Blue	1 White	Blue	2 White
2	Center	Orange	1 White	Orange	2 White
3	Center	Green	1 White	Green	2 White
4	First	Brown	1 White	Brown	2 White
5	First	Gray (Slate)	1 White	Gray (Slate)	2 White
6	First	Blue	1 Red	Blue	2 Red
7	First	Orange	1 Red	Orange	2 Red
8	First	Green	1 Red	Green	2 Red
9	First	Brown	1 Red	Brown	2 Red
10	First	Gray (Slate)	1 Red	Gray (Slate)	2 Red
11	First	Blue	1 Black	Blue	2 Black
12	First	Orange	1 Black	Orange	2 Black
13	Second	Green	1 Black	Green	2 Black
14	Second	Brown	1 Black	Brown	2 Black
15	Second	Gray (Slate)	1 Black	Gray (Slate)	2 Black
16	Second	Blue	1 Yellow	Blue	2 Yellow
17	Second	Orange	1 Yellow	Orange	2 Yellow
18	Second	Green	1 Yellow	Green	2 Yellow
19	Second	Brown	1 Yellow	Brown	2 Yellow
20	Second	Gray (Slate)	1 Yellow	Gray (Slate)	2 Yellow
21	Second	Blue	1 Violet	Blue	2 Violet
22	Second	Orange	1 Violet	Orange	2 Violet
23	Second	Green	1 Violet	Green	2 Violet
24	Second	Brown	1 Violet	Brown	2 Violet
25	Second	Gray (Slate)	1 Violet	Gray (Slate)	2 Violet
26	Second	Red	1 White	Red	2 White

3.5 Pair twist.- The insulated conductors shall be twisted into pairs.

3.6 Cabling.- The twisted pairs shall be arranged in layers, one pair thick, to form a cylindrical core. The number of different twist lengths alternating in a layer shall be not less than three, consistent with the lay up of the cable. Adjacent layers shall contain pairs of different lengths of pair twist. The center layer for Type I cables shall consist of 3 pairs; the 1st layer 9 pairs, and the 2nd layer 14 pairs; the twisted pairs shall be cabled with a left hand lay of not more than 5.0 inches. The center layer for Types II and III cables shall consist of 4 pairs; and the outer layer 10 pairs; the twisted pairs shall be cabled with a left hand lay of not more than 5.0 inches.

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3.7 Marker.- Manufacturer's markers of nonhygroscopic material denoting the manufacturer shall be laid parallel under the binder of the core or under the core wrap if no binder is used. Standard commercial identification marker threads may be used among the twisted pairs. The yarn used for marker threads shall be glass yarn, silk, or rayon.

3.8 Binder.- An open binder of nylon yarn, may be applied over the core with a right hand lay if necessary in the production process.

3.9 Core wrap.- The core shall be covered with a 0.001 inch Mylar tape or equal which shall be applied helically with an overlap to provide at least two thickness of material over the core. As an alternative procedure, two layers of 0.001 inch Mylar tape may be used with each layer having an overlap of not less than 15 percent. Both layers shall be applied helically in either the same or opposite directions.

3.10 Jacketing.- A tightly fitting seamless jacket of black polyvinylchloride shall be applied over the core wrap. The nominal jacket thickness shall be 0.050 inch and the minimum thickness shall be not less than 0.045 inches. The diameter of the finished Type I cable shall not be greater than 0.550 inches. The diameters of the finished Type II and Type III cables shall not be greater than 0.450 inches. The polyvinylchloride jacketing compound shall be as specified in 3.10.1.

3.10.1 Polyvinylchloride jacketing compound.- A polyvinylchloride jacketing compound such as Geon 8677 as made by B.F. Goodrich, or equal; Compound No.NF-3020 as made by Nougatuck Chemical Division of US Rubber Company, or equal, or Compound QFD-9924, Black 9975, as made by Plastic Division, Union Carbide Corporation, or equal, or Compound Koninor W-919 as made by Pantasote Company, or equal, shall be used.

3.11 Physical requirements of jacket before aging.- The jacket shall meet the requirements that follow when tested in accordance with 4.8.

3.11.1 Tensile strength.- The jacket specimen shall have a tensile strength of not less than 2000 pounds per square inch of original cross-sectional area at rupture. (See 4.8.1).

3.11.2 Elongation.- The jacket specimen shall have an elongation of the 2-inch gauge marks of not less than 7 inches at rupture. (See 4.8.2).

3.11.3 Tubing of jacket.- The specimen shall show no evidence of cracking of the jacket after testing in accordance with 4.8.3.

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3.12 Physical requirements of jacket after aging.- The jacket shall meet the requirements that follow when tested in accordance with 4.9.

3.12.1 Tensile strength.- The aged jacket specimen shall have a tensile strength not less than 90 percent of the value obtained on an adjacent specimen prior to artificial aging. (See 4.9.1).

3.12.2 Elongation.- The aged jacket specimen shall have an elongation of the 2-inch gauge marks, at rupture, not less than 90 percent of the value obtained on an adjacent specimen prior to artificial aging. (See 4.9.2).

3.13 Electrical requirements.-

3.13.1 DC resistance.- The direct current resistance of each conductor shall not exceed 26 ohms per 1000 ft. at 20°C. (See 4.10).

3.13.2 Dielectric strength.- The insulation shall withstand a DC potential of 3.0 kilovolts or an AC potential of 2100 volts rms applied between each conductor and the remaining conductors connected together. The potential shall be applied for a minimum of 3 seconds. (See 4.11).

3.13.3 Insulation resistance.- Immediately following the dielectric strength test, each conductor in turn measured against all other conductors shall have an insulation resistance of not less than 10,000 megohms per 1000 feet at 15.6°C. (See 4.12).

3.13.4 Mutual capacitance.- In each length, the mutual capacitance of each pair shall not exceed 0.090 microfarads per mile, when tested in accordance with 4.13.

3.13.5 Capacitance unbalance.- In every length, the 1000 cycle pair-to-pair capacitance unbalance shall not exceed 160 micro-microfarads for any length of 1000 feet or less. The average pair-to-pair capacitance unbalance shall not exceed 40 micro-microfarads for any length of 1000 feet or less. For other lengths, the values shall not exceed the value obtained by multiplying the value for 1000 feet by the square root of the ratio of the length in question to 1000 feet. (See 4.14).

3.13.6 Attenuation.- The attenuation at 1000 cycles per second at 20°C shall be not greater than 2.7 db per mile when measured in a mile length terminated with 600 ohms of resistance or when computed from measured primary cable characteristics: resistance, inductance, conductance, and capacitance. (See 4.15).

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3.14 Repairs.- No repair splices shall be made in the finished cable as a whole. Repairs made in the cable components during manufacture shall conform to the following requirements. (See 4.16).

3.14.1 Conductor strand splices.- When splices are required in the individual copper strands, they shall be of the butt-brazed type using silver solder. If the brazed splice is larger than the original wire, it shall be carefully dressed down to proper size which shall not be less than that of the original wire. Seven strand splices may be made in the stranding operation by joining the individual strands as described above provided that the individual strand splices are spaced at least one foot apart. No strands shall be kinked, and the smoothness of lay shall not be altered. When a splice is made in this manner, it shall be possible to move a die having a diameter of 0.032 inch and a length of one inch along the length of the spliced conductor without any binding or necessity for the exertion of a definite pressure.

3.14.2 Insulation repairs.- All insulation repairs shall be made with a suitable heat stabilized, non-oxidizing polyethylene compound and shall be of the heat molded type. In making such repairs, the original insulation shall be removed by smooth tapering cuts, extending on each side of the defect without nicking the conductor. The finished repair shall be cylindrical and smooth with no flash of excess patching compound. The minimum wall thickness requirements of paragraph 3.3, shall be maintained as closely as possible. The maximum overall length of a repair shall be 4 inches.

3.15 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein: (See 4.5).

3.1	Construction
3.2	Conductors
3.3	Conductor insulation
3.4	Color code
3.5	Pair twist
3.6	Cabling
3.7	Marker
3.8	Binder
3.9	Core wrap
3.10	Jacketing
3.14	Repairs

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contra t or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Inspection covered by subsidiary documents. (See 4.3).

(b) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.4).

(2) Quality conformance inspection of preparation for delivery. (See 4.17).

4.3 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where required
Conductors	3.2

4.4 Quality conformance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements shall perform the inspection specified in 4.3 and 4.4.1 through 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion --may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. (See 6.4). Test equipment for Government verification inspection shall be made available by the contractor.

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4.4.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the inspection procedures of MIL-STD-105 using the ordinary inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical examination	3.15	4.5	1.0%	4.0%
Electrical				
DC resistance	3.13.1	4.10)	1.0%	
Dielectric strength	3.13.2	4.11)	for the	
Insulation resistance	3.13.3	4.12)	group	

4.4.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

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Table III - Group B inspection

Inspection	Req. Para.	Insp. Para.
Physical requirements of insulation before aging	3.3.1	4.6
Elongation	3.3.1.1	4.6.1
Tensile strength	3.3.1.2	4.6.2
Compression strength	3.3.1.3	4.6.3
Tubing of insulation	3.3.1.4	4.6.4
Deformation of insulation	3.3.1.5	4.6.5
Low Temperature insulation	3.3.1.6	4.6.6
Density and melting index of insulation	3.3.1.7	4.6.7
Physical requirements of jacket before aging	3.11	4.8
Tensile strength	3.11.1	4.8.1
Elongation	3.11.2	4.8.2
Tubing of jacket	3.11.3	4.8.3
Mutual capacitance	3.13.4	4.13
Capacitance unbalance	3.13.5	4.14
Attenuation	3.13.6	4.15

4.4.3 Group C inspection.- This inspection shall be as listed in Table IV, and shall be performed on sample units that have been subjected to and met group A and group B inspection.

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Table IV - Group C inspection

Inspection	Req. Para.	Insp. Para.
Physical requirements of insulation after aging	3.3.2	4.7
Elongation	3.3.2.1	4.7.1
Tensile strength	3.3.2.2	4.7.2
Compression strength	3.3.2.3	4.7.3
Thermal stress cracking	3.3.2.4	4.7.4
Physical requirements of jacket after aging	3.12	4.9
Tensile strength	3.12.1	4.9.1
Elongation	3.12.2	4.9.2
Repairs	3.14 thru 3.14.2	4.16

4.4.3.1 Sampling for inspection.- Six specimens for each group C inspection shall be selected per 10,000 feet or fraction thereof of finished cable on order, without regard to their quality, except that sample units inspected at the start of the contract shall be from the first cable produced.

4.4.3.2 Noncompliance.- The contractor shall immediately report, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective. (See 6.3).

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4.4.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all damage.

4.5 Visual and mechanical examination.- The cables shall be examined to determine compliance with the requirements of 3.15, Workmanship.

4.6 Physical requirements of insulation before aging.- The following tests shall be performed on the insulation removed from conductors prior to artificial aging. The tests shall be performed at a temperature of 18° to 32°C.(See 3.3.1).

4.6.1 Elongation.- The insulation marked with 1 inch gauge marks shall be tested for elongation in accordance with Method 3031 of J-C-98. The rate of elongation shall be not less than 2 inches per minute. At rupture, the specimen shall meet the requirements of 3.3.1.1. The test shall be repeated for 2 inch gauge marks.

4.6.2 Tensile strength.- The insulation shall be tested for tensile strength in accordance with Method 3021 of J-C-98. At rupture, the specimen shall meet the requirements of 3.3.1.2.

4.6.3 Compression strength.- The insulation shall be tested for compliance with the requirement of 3.3.1.3, as follows: The specimen shall be at least 2-1/2 inches in length and shall be marked with 2 inch gauge marks prior to the removal of the conductor strands. After removing the conductor strands, the specimen shall be slipped on a steel mandrel 0.018 $\pm$ 0.001 inch in diameter. Before the performance of the test, the specimen shall be so adjusted that no twisting or bunching of the insulation occurs and that the gauge marks remain exactly 2 inches apart. The specimen shall then be placed longitudinally between the parallel faces of 2 steel plates, the face of one being exactly 2 inches in length and the face of the other being at least 2 inches in length. A load resulting from decreasing the separation of the steel plates shall be applied to the specimen between the plates until the insulation crushes. The speed of the compression testing machine shall be such that when no specimen is between the steel plates, the rate of approach of the plates will be between 1.5 and 2.25 inches per minute. The compression strength shall be taken as that load which produces the initial abrupt reduction in the separation of the plates without proportionate increase in load, as indicated on a record chart of load versus thickness. The machine used for this test shall be a spring-loaded type H. L. Scott Compression Tester, or equal.

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4.6.4 Tubing of insulation.- To determine whether the extrusion process has caused strains in the insulation, a specimen of the insulated conductor shall be looped back and then wound tightly on itself for 5 close turns, the ends securely taped, and the specimen placed in an oven maintained at a temperature of 121° ± 1°C for 1 hour. After removal from the oven, the specimen shall be examined for compliance with 3.3.1.4 and shall be considered to have failed if there is any evidence of cracking of the insulation when examined under a magnifying glass of approximately 3 diameter magnification (focal distance 3 inches).

4.6.5 Deformation of insulation.- A Randall and Stickney Thickness Gauge, or equal, equipped with a 3/8 inch diameter circular gauge foot and with a 250 gram weight on the plunger shall be placed in an oven maintained at a temperature of 121° ± 1°C for 1 hour. A specimen of insulated conductor at room temperature shall then be placed on the gauge platform located centrally under the gauge foot, and its diameter read exactly 5 seconds after being so placed. The gauge and the specimen shall be allowed to remain in the oven, at the temperature specified above, for 1 hour and the diameter of the specimen read again without disturbing the specimen or gauge. The decrease in diameter during this period shall conform to 3.3.1.5.

4.6.6 Low temperature, insulation.- A specimen of insulated conductor and a test mandrel whose diameter shall be 0.0625 ± 0.003 inch shall be placed in a cold chamber maintained at a temperature of -55° ± 2°C for 24 hours. At the end of this time, and while still in the cold chamber, the insulated conductor shall be capable of being wound around the mandrel for 7 close turns at a rate of approximately 1 turn per second without visible evidence of cracking of the insulation when examined under a magnifying glass of approximately 3 diameter magnification (focal distance 3 inches). In conducting this test, the first and last turns shall be discounted and no object having a temperature higher than -55° ± 2°C shall come within 12 inches of the part of the specimen which is under test. (See 3.3.1.6).

4.6.7 Density and melt index of insulation.- The insulation shall be inspected for compliance with 3.3.1.7.

4.7 Physical requirements of insulation after aging.- The insulation removed from the conductors before cabling shall be tested after the samples have been placed in an air oven conforming to the requirements of ASTM Standard D573, with the temperature maintained at 100° ± 1°C for a period of 7 days. The samples shall then be cooled for a minimum of 24 hours at room temperature and tested at a temperature of 18° to 32°C as follows: (See 3.3.2).

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4.7.1 Elongation.- The insulation marked with 2 inch gauge marks shall be tested for elongation in accordance with Method 3031 of J-C-98. The specimen shall meet the requirements of 3.3.2.1.

4.7.2 Tensile strength.- The insulation shall be tested for tensile strength in accordance with Method 3021 of J-C-98. At rupture, the specimen shall meet the requirements of 3.3.2.2.

4.7.3 Compression strength.- The insulation shall be tested for compliance with the requirement of 3.3.2.3 in the manner specified in 4.6.3.

4.7.4 Thermal stress cracking.- To determine whether the insulation is free of strain which will cause subsequent cracking, a specimen shall be looped back and then wound tightly on itself for 5 close turns, the ends securely taped and the specimen placed in an oven at a temperature of 100° ± 1°C for 14 days. After removal from the oven and unwrapped, the specimen shall be considered to have failed if there is any evidence of cracking of the insulation when examined under a magnifying glass of approximately 3 diameter magnification (focal distance 3 inches). (See 3.3.2.4).

4.8 Physical requirements of jacket before aging.- The following tests shall be performed with the jacket sections removed from the cable or taken from a slab section, whichever is practicable, at a temperature of 18° to 32°C.(See 3.11)

4.8.1 Tensile strength.- The jacket shall be tested for tensile strength in accordance with Method 3021 of J-C-98. At rupture, the specimen shall meet the requirements of 3.11.1.

4.8.2 Elongation.- The jacket marked with 2 inch gauge marks shall be tested for elongation in accordance with Method 3031 of J-C-98. At rupture, the specimen shall meet the requirements of 3.11.2.

4.8.3 Tubing of jacket.- To determine whether the extrusion process has caused strains in the jacket, a specimen of the jacketed cable shall be wound tightly on a mandrel, having a diameter of 0.625 ± 0.005 inch, for 5 close turns, the ends securely anchored and the specimen placed in an oven maintained at a temperature of 120° ± 1°C, for 1 hour. After removal from the oven the specimen shall be considered to have failed if there is any evidence of cracking of the jacket when examined under a magnifying glass of approximately 3 diameter magnification (focal distance 3 inches). (See 3.11.3).

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4.9 Physical requirements of jacket after aging.- The jacket sections removed from the cable or taken from a slab section, whichever is practicable, shall meet the following tests when tested at a temperature of 18° to 32°C, after aging as follows: Place the sample in the air oven conforming to ASTM Standard D573 and maintain at a temperature of 80° ± 1°C for a period of 14 days, and follow with a cooling period of at least 24 hours at room temperature. (See 3.12).

4.9.1 Tensile strength.- The jacket shall be tested for tensile strength in accordance with Method 3021 of J-C-98. The specimen shall meet the requirements of 3.12.1.

4.9.2 Elongation.- The jacket marked with 2 inch gauge marks shall be tested for elongation in accordance with Method 3031 of J-C-98. At rupture, the specimen shall meet the requirements of 3.12.2.

4.10 DC resistance.- The dc resistance of each pair of cable length shall be measured in accordance with Method 6021 of J-C-98 for compliance with 3.13.1.

4.11 Dielectric strength.- The cables shall be tested for compliance with the requirements of 3.13.2.

4.12 Insulation resistance.- The insulation resistance of the cables shall be measured in accordance with Method 6031 of J-C-98 for compliance with 3.13.3. In making the insulation resistance test, an electro-motive force of not less than 100 volts dc shall be used, the conductors being maintained negative with respect to ground. The insulation resistance shall be computed from the galvanometer deflection obtained after an electrification of 1 minute. In making the insulation resistance test, the test may be terminated in less than 1 minute if the galvanometer has ceased fluctuating and the reading indicates the required minimum insulation resistance has been obtained. However, test results on 5 percent of the lengths after 1 minute electrification shall be recorded to permit continuous check on quality.

4.13 Mutual capacitance.- The cable length shall be tested for compliance with the requirements of paragraph 3.13.4 at a temperature of 15.6°C and at a frequency of 1000 ± 100 cycles per second. The mutual capacitance of a pair is that measured between the 2 wires of a pair and the remainder of the conductors grounded. One pair from the center layer, 2 pairs from the first layer and 3 pairs from the second layer shall be selected at random and tested for mutual capacitance.

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4.14 Capacitance unbalance.- The cable length shall be tested for compliance with the requirements of paragraph 3.13.5. Capacitance unbalance shall be measured between each pair and its adjacent pairs in each layer, and between each pair in a layer and all others pairs in adjacent layers.

4.15 Attenuation.- The cable length shall be tested for compliance with the requirements of paragraph 3.13.6. One pair from the center layer, 2 pairs from the first layer, and 3 pairs from the second layer shall be selected at random and tested for attenuation.

4.16 Repairs.- Repairs shall be inspected for compliance with the requirements of 3.14 to 3.14.2 inclusive.

4.17 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking of telephone cables shall be in accordance with MIL-C-12000.

6. NOTES

6.1 Intended use.- Types I, II, and III telephone cables are intended for field communications system.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, and level C).

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6.3 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.4.3.2).

6.4 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN

ARMY EL

Preparing Activity

ARMY EL

Project No. 6145-A004

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DEPARTMENT OF THE ARMY
HEADQUARTERS
U.S.ARMY ELECTRONICS COMMAND
FORT MONMOUTH, NEW JERSEY 07703

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding General
U. S. Army Electronics Command
ATTN: AMSEL-PP-
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity).*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

To detach this form, cut along this line

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ESC-FM 2011-66

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